Tianrui (Dennis) Wu

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EDUCATION

University of Arizona — **Tucson, AZ**
Bachelor of Science in Physics and Mathematics — *Jan 2026 – May 2027 (Expected)*
- ***GPA:*** 3.87/4.00, Dean's List, **Global Wildcat Scholarship** *($10k/year merit-based)*
- ***Coursework:*** Stochastic Calculus, Statistical Machine Learning, Advanced Mathematics Modeling, Computational Physics

Duke University — **Durham, NC**
Bachelor of Science in Physics and Mathematics, Computer Science Minor — *Aug 2021 – Dec 2024*
- ***GPA:*** 3.92/4.00, Dean's List *with distinction*, **Faculty Scholars Finalist 24'** (*highest honor, sole physics nominee*)
- ***Selected Coursework:*** Real Analysis, Advanced Probability, Stat. Learning & Inference, Graduate ML, Game Theory
- ***Extracurriculars:*** **Math TA (lin. algebra., prob.)**, Duke University; **Founder & President**, Duke Astronomy Club; **Chapter President**, Society of Physics Students; **APS Ambassador**; **Senior Editor**, Duke Vertices Science Journal

HONORS & AWARDS

Olympiads:
- National 1st Place, *US Physicists' Tournament 2026*
- Top Gold, *British Physics Olympiad 2021 (top 10)*
- Third Grand Award, *Regeneron ISEF 2021 (top 10)*

Scholarships:
- J. B. Burnell Scholarship 2024 *(sole recipient)*
- Member, ΣΠΣ *Honor Society (reserved for top physicists)*
- Outstanding Chapter Award 2023, *Society of Physics Students*

EXPERIENCE

Qrigin Capital *(start-up prop shop, $35M)* — **Remote**
Quantitative Researcher, Systematic Trader — *Aug 2025 – Present*
- ***Alpha Discovery***: Built an LSTM forecasting engine on phase-space features with full MLOps (MLflow), integrated 200+ multi-frequency factors with decay analysis and cross-sectional ranking, lifting win rates 3.1%
- ***Directional Strategy:*** Developed HFT taker strategy with tick-data, capturing short reversions with confidence signal, deployed live at post-fee Sharpe 8.2 over 3 months on Coinbase Intl
- ***Optimal Execution***: Architected execution algorithms for CTA books using microstructure timing overlays modulating child-order aggressiveness, cutting slippage ~2 bps vs TWAP across major exchanges
- ***Market Making:*** Built a dual-engine tick-based backtest stack with vectorized L2 queue-position fills, developed and validated a grid-based market-making strategy with out-of-sample Sharpe 9 for over 3 months
- ***Trading Operations***: Directed tuning model parameters to market conditions and strategy factors while owning daily operations like P&L monitoring, TCA, fee analysis; monitoring live risk, trimmed exposure ahead of October crash

Applied Technologies: Python (*PyTorch, MLflow, Polars, pandas, NumPy*), CEX APIs (*REST/WebSocket*), Linux, Git

InsightCheck Investments *(systematic equity fund with CTA desk, $300M AUM)* — **Shenzhen, China**
Quantitative Researcher — *Feb 2025 – Aug 2025*
- ***Statistical Arb:*** Enhanced quantitative models across equity/futures markets and developed market-neutral L/S equity strategy combining PCA statistical factors with ML signals & covariance-based risk model, backtested with post-cost Sharpe of 2.4.
- ***Factor Research:*** Researched fundamental and microstructure-based alpha factors for Chinese A-shares, validating 62+ intraday factors with daily IC >0.01 and conducting cross-sectional portfolio optimization in Polars.
- ***Orderbook Engine & Infra***: Solely engineered high-performance orderbook reconstruction engine using L3 market-by-order data, prototyping in Python and migrating to C++, achieving ~1 second-per-day process time of full A-share universe.

Applied Technologies: Python (*Polars*), C++, SQL, Linux, Git

PROJECTS

Density-field Forecasting with High Redshift Uncertainty | Python (*pandas, NumPy, Numba*), Linux HPC — *2026*
- Led a first-author cosmological research project as sole analyst, coordinating with a PI and postdoctoral researcher
- Diagnosed and corrected a systematic bias in a Bayesian MCMC inference pipeline via cross-validated covariance estimation across sensitivity grid, cutting the bias 6-fold. First-author paper in prep: *BAO Reconstruction with Photo-z, PRD (2026)*

IMC Prosperity 4 Trading Competition | Python (*PyTorch*), Rust — *2026*
- Ranked top 3% of 6,000+ teams using market-making and cross-asset arbitrage driven by fingerprinting of counterparties.

Crypto Market Maker with Backtest Engine | Python (*pandas, Polars, NumPy, Numba, PyTorch*), Rust — *2025*
- Built and deployed live ETH market making framework on OKX, combining EWMA flow alphas with CNN capturing orderbook patterns via phase-space feature rep., averaged 0.5-1% daily return. *Project led to recruitment at Qrigin*

Neural Network Prediction of Power Spectrum Window Function | Python (*PyTorch*), Linux HPC — *2024*
- As Caltech SURF Research Fellow with SPHEREx team, built a ResUNet CNN (PyTorch) for signal deconvolution on large-scale data, owning mock-data generation, distributed HPC training, achieved sub-percentage error

Understanding Dark Matter with Data | Python, Linux HPC — *2022-2024*
- As a research assistant at Duke Cosmology Group, built Python pipelines analyzing 120 TB of simulation to extract galaxy features, applying MCMC-based Bayesian inference, published papers: *[JWST], ApJ (2024); [OpenUniverse], MNRAS (2025)*

SKILLS

Skills: Python *(PyTorch, Polars, pandas, NumPy, Numba, Dask, MLflow)*, Rust, C/C++, *Postgre*SQL, Unix
Markets: Digital Assets, Commodity Futures, Global Equities, Options (*completed Akuna 101 & 201*), Prediction Markets
Interests: Poker, Ranked GeoGuessr, Table Wargaming, Strategy Games, Chaos Theory, Forecasting, Astrophotography, Sci-fi